

Notes: Loh and Stulz (RFS, 2011)

When Are Analyst Recommendation Changes Influential?

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1 Big picture

Question. When do sell-side analyst recommendation changes visibly affect an individual firm's stock price or trading activity, rather than merely producing a small average event-study effect across many firms?

Setting. The paper studies U.S. analyst recommendation changes from I/B/E/S over 1994–2006, with recommendation dates improved using First Call when possible. Recommendations are coded on a five-point scale from sell to strong buy, and a recommendation change is the analyst's current rating minus the analyst's own prior outstanding rating.

Core challenge. The analyst literature usually asks whether recommendation changes have significant *average* abnormal returns. This paper argues that average effects do not answer whether an individual recommendation is noticeable to investors following a particular stock. A one-percent downgrade reaction can be statistically significant in a large sample but indistinguishable from noise for a single firm.

Main conceptual contribution. Loh and Stulz define an *influential recommendation change* at the firm-event level. A recommendation is influential only if the event return is both in the correct direction and statistically large relative to the firm's own past idiosyncratic volatility. They also use abnormal turnover as a separate definition, because an analyst may affect trading even when the price reaction is not large enough to pass the return threshold.

Main empirical design. The paper first removes recommendation days likely contaminated by firm-specific news: days around earnings announcements, days around management earnings guidance, and firm-days with multiple analyst recommendations. The authors then classify the remaining recommendation changes as influential or non-influential and compare their analyst, recommendation, firm, and post-event characteristics.

Main baseline result. Only about 11.7% of recommendation changes are influential in returns and about 12.8% are influential in turnover. Thus, most recommendation changes are not visibly influential, but the influential subset is far larger than the chance benchmark implied by a five-percent statistical threshold.

Key analyst result. Recommendation changes are more likely to be influential when they are issued by analysts who were influential before, star analysts, leader analysts, or analysts issuing the recommendation together with an earnings forecast.

Key recommendation result. Recommendations that move away from the consensus are more likely to be influential. This is consistent with the idea that influential recommendations often challenge the market's current view of the firm rather than merely updating a common view.

Key firm result. Influential recommendations are more likely for growth firms, smaller firms, firms with high institutional ownership, firms with high forecast dispersion, and firms with lower analyst activity. These are environments where the market's valuation model may be less settled or where informed investors are more likely to pay attention.

Key mechanism result. Influential recommendations are followed by more analyst activity, higher stock turnover, higher volatility, larger consensus forecast revisions, and sometimes large same-direction industry returns. This supports the paper's "paradigm shift" interpretation: a strong analyst call can change how investors and other analysts view a firm.

Economic takeaway. Analysts do not move prices most of the time. Their value is concentrated in a relatively small set of visible, high-impact calls. Studying averages alone understates this hetero-

generality and misses the distinction between routine recommendation changes and recommendation changes that actually reshape beliefs.

2 Data and measurement (key objects)

2.1 Recommendation data and sample construction

The paper uses the I/B/E/S Detail U.S. recommendation file for recommendation changes from 1994 to 2006. Prior ratings can come from 1993, but current ratings in the main sample start in 1994.

A rating is treated as outstanding only if it has been reviewed within the past twelve months and has not been stopped by the broker. The paper excludes initiations, anonymous analysts, observations with no prior outstanding rating, low-priced stocks with lagged price below one dollar on event day 0, and observations with fewer than three analysts contributing to the outstanding consensus.

Date adjustment. I/B/E/S recommendation dates are matched to First Call dates by broker, firm, and recommendation level. When a match is found, the First Call date is used. This matters because daily event studies are sensitive to whether the recommendation date is off by one day.

Main descriptive sample. The full descriptive sample contains 154,134 recommendation changes. The most common categories are one-point downgrades, one-point upgrades, reiterations, and two-point downgrades.

Interpretation. The sample is designed to capture actual changes in analysts' stated views, not first-time coverage initiations or stale recommendation records.

2.2 Recommendation scale and recommendation change

Recommendations are coded as

$$1 = \text{Sell}, \quad 2 = \text{Underperform}, \quad 3 = \text{Hold}, \quad 4 = \text{Buy}, \quad 5 = \text{Strong Buy}. \quad (1)$$

The recommendation change is

$$\text{Recchg}_i = \text{Current recommendation}_i - \text{Prior recommendation}_i. \quad (2)$$

Thus, $\text{Recchg}_i > 0$ is an upgrade and $\text{Recchg}_i < 0$ is a downgrade. Reiterations have $\text{Recchg}_i = 0$.

Interpretation. The sign of Recchg_i determines the economically correct direction of the stock-price reaction: upgrades should have positive abnormal returns; downgrades should have negative abnormal returns.

2.3 Event abnormal return

The paper uses a two-day event window, days $[0, 1]$, because most of the recommendation-related return occurs on the event day and the next trading day. Day 0 is the First Call adjusted recommendation date, or the next trading day when the recommendation is issued after the market close or on a non-trading day.

For recommendation change i , the two-day buy-and-hold abnormal return is

$$\text{CAR}_i = \prod_{t=0}^1 (1 + R_{it}) - \prod_{t=0}^1 (1 + R_{it}^{\text{DGTW}}). \quad (3)$$

Here R_{it} is the raw return of the stock, and R_{it}^{DGTW} is the return on a Daniel-Grinblatt-Titman-Wermers benchmark portfolio matched on size, book-to-market, and momentum.

Interpretation. The benchmark removes common return components associated with stock characteristics. The remaining CAR_i is the event-specific abnormal return around the recommendation.

2.4 Firm-news contamination screens

The paper stresses that analysts often issue recommendation changes when firms release news. If those observations are kept, the researcher may wrongly attribute the firm’s own news to the analyst.

The main clean sample removes recommendation changes that occur:

- in a three-day window around the firm’s quarterly earnings announcement;
- in a three-day window around management earnings guidance;
- on firm-days when multiple analysts issue recommendations.

Interpretation. This is the paper’s central credibility step. It does not remove every possible news event, but it removes the most systematic forms of firm-news contamination.

2.5 Influential in returns

A recommendation change is influential in returns if it satisfies two conditions:

$$\text{sign}(\text{CAR}_i) = \text{sign}(\text{Recchg}_i), \quad (4)$$

$$|\text{CAR}_i| > 1.96\sqrt{2}\sigma_{\varepsilon,i}. \quad (5)$$

The idiosyncratic volatility $\sigma_{\varepsilon,i}$ is estimated from a daily time-series regression of the firm’s returns on the market return and the Fama-French factors SMB and HML over trading days $[-69, -6]$ before the recommendation date.

Interpretation. This is a firm-level event-study significance test. The recommendation is influential only if the price move is large enough to be visible relative to the firm’s own recent residual volatility.

2.6 Influential in turnover

The turnover definition focuses on whether investors trade unusually heavily after the recommendation. Daily abnormal turnover is defined as

$$\text{abturn}_{it} = \log(\text{turnover}_{it} + 0.00000255) - \overline{\log(\text{turnover}_i + 0.00000255)}, \quad (6)$$

where the bar denotes the average over the prior three months.

The recommendation is influential in turnover if cumulative abnormal turnover over days $[0, 1]$ is unusually high:

$$\sum_{t=0}^1 \text{abturn}_{it} > 1.96\sqrt{2}\sigma_{\text{abturn},i}, \quad (7)$$

where $\sigma_{\text{abturn},i}$ is the standard deviation of abnormal turnover over days $[-69, -6]$.

Interpretation. The turnover rule captures recommendation changes that are noticed and acted on by investors, even if the price reaction itself does not pass the return threshold.

2.7 Analyst and recommendation characteristics

The paper studies several analyst-level and recommendation-level variables.

Forecast accuracy. Analysts are ranked into firm-year accuracy quintiles based on their last unrevised FY1 earnings forecast. Lower ranks mean greater accuracy.

Away from consensus. This dummy equals one if the new recommendation is farther from the current consensus than the prior recommendation.

Star analyst. This dummy equals one if the analyst is ranked in the Institutional Investor All-American poll.

Analyst experience. The paper uses absolute experience in I/B/E/S and relative firm-specific experience compared with other analysts covering the same stock.

Concurrent earnings forecast. This dummy equals one if the analyst issues any earnings forecast in the three-day window around the recommendation change.

Influential before. The paper measures whether the analyst had an influential recommendation in the past, either for any stock or for the same stock.

Leader-follower ratio. The leader-follower ratio measures whether other analysts quickly follow a recommendation:

$$\text{LFR}_i = \frac{\text{gap sum between the current recommendation and the previous two recommendations by other brokers}}{\text{gap sum between the current recommendation and the next two recommendations by other brokers}}. \quad (8)$$

Interpretation. An LFR greater than one indicates that the analyst is a leader whose recommendation is quickly followed by other brokers.

2.8 Firm characteristics

The main firm variables include book-to-market, size, prior twelve-month momentum, institutional ownership, turnover, idiosyncratic volatility, total volatility, earnings forecast dispersion, and analyst activity measured by the number of earnings forecasts issued in the prior three months.

Interpretation. These variables capture whether the firm is easy or hard to value, whether investors are likely to pay attention, and whether the analyst is speaking into a crowded information environment.

3 Models and empirical specifications (all equations + interpretation)

3.1 Average event-study benchmark

Before defining influential recommendations, the paper estimates the average event CAR by recommendation-change category:

$$\overline{\text{CAR}}_k = \frac{1}{N_k} \sum_{i:\text{Recchg}_i=k} \text{CAR}_i, \quad (9)$$

where $k \in \{-4, -3, -2, -1, 0, +1, +2, +3, +4\}$.

Interpretation. This reproduces the traditional event-study approach. It asks whether the average downgrade or upgrade return is statistically significant, but it does not tell whether any individual recommendation visibly moved a stock.

3.2 Main return-based influential indicator

The main binary outcome is

$$\text{InfluentialReturn}_i = \mathbf{1} \left\{ \text{sign}(\text{CAR}_i) = \text{sign}(\text{Recchg}_i) \text{ and } |\text{CAR}_i| > 1.96\sqrt{2}\sigma_{\varepsilon,i} \right\}. \quad (10)$$

Interpretation. The definition requires both economic direction and statistical visibility. A large wrong-signed return does not count.

3.3 Turnover-based influential indicator

The turnover-based outcome is

$$\text{InfluentialTurnover}_i = \mathbf{1} \left\{ \sum_{t=0}^1 \text{abturn}_{it} > 1.96\sqrt{2}\sigma_{\text{abturn},i} \right\}. \quad (11)$$

Interpretation. This definition does not require a positive or negative price direction. It asks whether the recommendation generated abnormal investor trading.

3.4 Predictive probit specification

The central multivariate model is a probit:

$$\Pr(\text{Influential}_i = 1) = \Phi \left(\alpha + X_i^{\text{rec}}\beta + X_i^{\text{analyst}}\gamma + X_i^{\text{firm}}\delta + \lambda_1 \text{RegFD}_i + \lambda_2 \text{Settlement}_i \right). \quad (12)$$

The dependent variable is either $\text{InfluentialReturn}_i$ or $\text{InfluentialTurnover}_i$.

The recommendation controls include recommendation level, absolute recommendation-change magnitude, and an upgrade dummy. Analyst controls include prior influence, forecast accuracy, away from consensus, star status, experience, concurrent forecast, and past leader-follower ratio. Firm controls include book-to-market, size, momentum, institutional ownership, turnover, idiosyncratic volatility, dispersion, and analyst activity.

Estimation detail. The paper reports probit coefficients and marginal effects. Standard errors are clustered in two dimensions, by analyst and by firm.

Interpretation. The probit asks which observable characteristics at the time of the recommendation predict whether the recommendation will be visibly influential.

3.5 Change in firm environment around the event

The paper compares changes in firm and analyst activity before and after the recommendation:

$$\Delta Z_i = Z_{i,\text{post}} - Z_{i,\text{pre}}. \quad (13)$$

For volatility and turnover, the paper compares the prior three months with the following three months, excluding the days around the event. For analyst activity and earnings forecast revisions, it compares analogous pre- and post-event windows.

Interpretation. If influential recommendations represent paradigm shifts, they should not only move prices at the event date; they should also change subsequent trading, volatility, analyst activity, and forecast revisions.

3.6 Robustness and alternative definitions

The paper checks whether the influential fraction is robust to alternative definitions:

- using the current recommendation minus the last recommendation by any analyst;
- using the current recommendation minus the latest consensus recommendation;
- using raw returns rather than DGTW abnormal returns;
- estimating prior volatility from recommendation-free days only;
- dropping observations with unusually large pre-event returns over days $[-2, -1]$.

Interpretation. The key result is not an artifact of a single recommendation-change definition, a particular benchmark, or a few pre-event news days.

4 Identification and credibility (what makes the estimates credible)

4.1 What is being identified

The paper does not claim to estimate a structural causal effect of analyst reports in every event. Instead, it identifies recommendation changes whose associated price or turnover movement is large enough to be visible at the stock level after the main sources of firm-news contamination are removed.

Interpretation. The goal is classification and prediction: which recommendation changes are visibly influential, and which analyst and firm conditions make this more likely?

4.2 Why average event studies are insufficient

Average abnormal returns can be statistically significant even when no individual recommendation produces a return large enough to be noticed for a given firm. Conversely, a small number of high-impact calls can drive an average effect.

The paper's firm-level rule addresses this by comparing each event return to the firm's own recent residual volatility.

Economic message. The paper shifts attention from the average analyst recommendation to the tail of recommendations that actually move beliefs.

4.3 Dealing with firm-specific news

The most serious concern is that analysts issue recommendation changes in response to the same firm news that moves the stock. The paper responds by removing earnings announcement windows, management guidance windows, and multiple-recommendation firm-days.

The paper shows that these screens matter. For one-point downgrades, the mean CAR falls from

−3.551% in the full sample to −1.562% after the main firm-news screens. For one-point upgrades, the mean CAR falls from 2.503% to 1.914%.

Interpretation. Firm news explains a large part of the average recommendation reaction, but it does not eliminate the existence of visible influential recommendation changes.

4.4 Date accuracy

The paper improves event-date accuracy by hand-matching I/B/E/S recommendations to First Call observations. About 52% of I/B/E/S observations are matched to First Call, and the First Call date is used when available.

Interpretation. This lowers the risk that price reactions are missed or assigned to the wrong day because the recommendation date is off by one trading day.

4.5 Residual confounding and limits

Some uncaptured corporate news may remain. The paper addresses this with an additional pre-event return screen and a random Factiva news check. Dropping observations with large pre-event returns leaves the influential-in-returns fraction at 11.1%, close to the baseline 11.7%.

Limit. The paper cannot fully prove that the analyst call alone caused every influential event. It can show that the influential subset survives major news screens and exhibits systematic analyst, recommendation, and firm characteristics.

4.6 Why the mechanism is plausible

Influential recommendations are associated with higher leader-follower ratios, more post-event analyst activity, higher turnover, higher volatility, larger forecast revisions, and large same-direction industry returns. These patterns are hard to reconcile with a purely random tail-event interpretation.

Interpretation. The post-event evidence supports the idea that influential recommendations trigger broader information processing by investors and analysts.

5 Tables and figures

5.1 Table 1 and Figure 1: recommendation-change distribution

Read:

- The main descriptive sample contains 154,134 recommendation changes.
- The largest category is one-point downgrades: 51,679 observations, or 33.5% of the sample.
- The second largest category is one-point upgrades: 44,498 observations, or 28.9% of the sample.
- Reiterations account for 21,331 observations, or 13.8%.
- Large changes of three or four rating points are rare.

Economic message. The empirical action is concentrated in modest one-point changes. The paper’s influential-recommendation idea is therefore mostly about identifying which ordinary-looking recommendation changes become high-impact events.

Table 1
Descriptive statistics of recommendation changes

Panel A: Transition probabilities of recommendation changes

Prior Rec	Current Rec					
	1 Sell	2 Underperform	3 Hold	4 Buy	5 Strong Buy	Total
1 (Sell)	90	130	2,008	253	223	2,704
	3.3%	4.8%	74.3%	9.4%	8.2%	100%
2 (Underperform)	125	1,079	5,191	801	160	7,356
	1.7%	14.7%	70.6%	10.9%	2.2%	100%
3 (Hold)	2,139	5,554	7,661	24,098	12,195	51,647
	4.1%	10.8%	14.8%	46.7%	23.6%	100%
4 (Buy)	333	1,065	30,406	8,093	15,079	54,976
	0.6%	1.9%	55.3%	14.7%	27.4%	100%
5 (Strong Buy)	371	338	16,740	15,594	4,408	37,451
	1.0%	0.9%	44.7%	41.6%	11.8%	100%
Total	3,058	8,166	62,006	48,839	32,065	154,134

Panel B: Recommendation change categories

Rec Change	Frequency	Percentage
-4	371	0.2%
-3	571	0.4%
-2	19,944	12.9%
-1	51,679	33.5%
0	21,331	13.8%
+1	44,498	28.9%
+2	15,004	9.7%
+3	413	0.3%
+4	223	0.1%
Total	154,134	100%

The sample of recommendation (rec) changes are from I/B/E/S Detail U.S. File 1994 to 2006. Each rec change (or rating change) is an analyst's current rating minus his prior rating (prior ratings may be from 1993, but current ratings are from 1994 onward). Analyst initiations or ratings with no prior outstanding ratings are excluded. A rating is outstanding if it has been confirmed by the analyst (in the I/B/E/S review date field) in the last twelve months and has not been stopped by the broker (in the I/B/E/S Stopped File). Ratings are coded as 1 (sell) to strong buy (5), and rating changes lie between -4 and 4. Anonymous analysts and observations with less than three analysts with outstanding ratings are excluded. Panel A reports the transition probabilities of rec changes. For example, in column 1, when the prior rec is a sell, it has a 4.8% probability of transitioning to an underperform rating. Panel B reports the frequencies of each rec change category.

Prior to Current
Transition Prob.

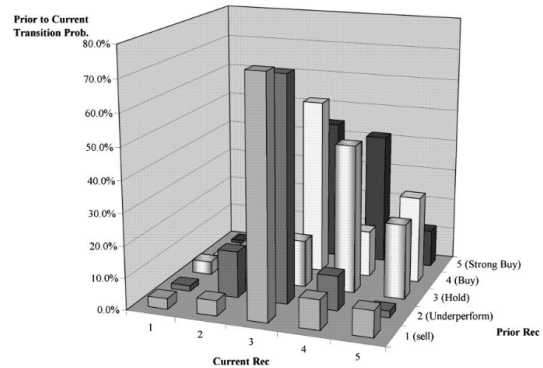


Figure 1
Transition probabilities of recommendation changes

The sample of recommendation (rec) changes is from I/B/E/S Detail U.S. File 1994 to 2006. Each rec change (or rating change) is an analyst's current rating minus his prior rating (prior ratings may be from 1993, but current ratings are from 1994 onward). Analyst initiations or ratings with no prior outstanding ratings are excluded. A rating is outstanding if it has been confirmed by the analyst (in the I/B/E/S review date field) in the last twelve months and has not been stopped by the broker (in the I/B/E/S Stopped File). Ratings are coded as 1 (sell) to strong buy (5), and rating changes lie between -4 and 4. Firms with less than three analysts making up the consensus are excluded. The chart plots the transition probabilities of rec changes—the probability that a prior rec transits to any of the five rating categories.

5.2 Table 2: filters and average CARs

Table 2
The impact of various filters on recommendation event percentage CAR

Filtered Samples	Mean	Std	% CAR > 0	N	R2 adj	Percentile				
						50%	75%	Median	25%	1%
Recommendation Change = +1										
1) Full sample	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
2) No earnings date	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
3) No earnings date or earnings date	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
4) No earnings date or earnings date or earnings date	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
5) Earnings 125 days before or after	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
6) Earnings 125 days before or after	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
Recommendation Change = +2										
1) Full sample	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
2) No earnings date	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
3) No earnings date or earnings date	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
4) No earnings date or earnings date or earnings date	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
5) Earnings 125 days before or after	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
6) Earnings 125 days before or after	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
Recommendation Change = +3										
1) Full sample	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
2) No earnings date	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
3) No earnings date or earnings date	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
4) No earnings date or earnings date or earnings date	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
5) Earnings 125 days before or after	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
6) Earnings 125 days before or after	-0.000**	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000

Table 3
Continued

Filtered Samples	Mean	Std	% CAR > 0	N	R2 adj	Percentile				
						50%	75%	Median	25%	1%
Recommendation Change = +4										
1) Full sample	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
2) No earnings date	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
3) No earnings date or earnings date	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
4) No earnings date or earnings date or earnings date	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
5) Earnings 125 days before or after	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
6) Earnings 125 days before or after	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
Recommendation Change = +5										
1) Full sample	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
2) No earnings date	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
3) No earnings date or earnings date	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
4) No earnings date or earnings date or earnings date	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
5) Earnings 125 days before or after	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
6) Earnings 125 days before or after	-0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
Recommendation Change = +2										
1) Full sample	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
2) No earnings date	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
3) No earnings date or earnings date	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
4) No earnings date or earnings date or earnings date	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
5) Earnings 125 days before or after	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
6) Earnings 125 days before or after	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
Recommendation Change = +1										
1) Full sample	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
2) No earnings date	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
3) No earnings date or earnings date	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
4) No earnings date or earnings date or earnings date	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
5) Earnings 125 days before or after	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
6) Earnings 125 days before or after	0.000	0.000	0.000	1,200	0.000	-0.000	-0.000	-0.000	-0.000	-0.000

Table 4
Continued

Filtered Samples	Mean	Std	% CAR > 0	N	R2 adj	Percentile				
						50%	75%	Median	25%	1%
1) Full sample	1.000**	0.000	0.000	1,200	0.000	0.000	0.000	0.000	0.000	
2) No earnings date	1.000**	0.000	0.000	1,200	0.000	0.000	0.000	0.000	0.000	
3) No earnings date or earnings date	1.000**	0.000	0.000	1,200	0.000	0.000	0.000	0.000	0.000	
4) No earnings date or earnings date or earnings date	1.000**	0.000	0.000	1,200	0.000	0.000	0.000	0.000	0.000	
5) Earnings 125 days before or after	1.000**	0.000	0.000	1,200	0.000	0.000	0.000	0.000	0.000	
6) Earnings 125 days before or after	1.000**	0.000	0.000	1,200	0.000	0.000	0.000	0.000	0.000	

The descriptive statistics of the abnormal returns to recommendation (R2) changes are reported. Recommendations are from I/B/E/S (2004) for FY 1999-2000. Rec data are from FY 1999-2000. CAR is calculated as the difference between the actual return and the expected return. The expected return is calculated as the return to the market portfolio. The abnormal return is calculated as the difference between the actual return and the expected return. The abnormal return is calculated as the difference between the actual return and the expected return. The abnormal return is calculated as the difference between the actual return and the expected return. The abnormal return is calculated as the difference between the actual return and the expected return. The abnormal return is calculated as the difference between the actual return and the expected return. The abnormal return is calculated as the difference between the actual return and the expected return. 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The sample of recommendation (rec) changes is from I/B/E/S Detail U.S. File 1994 to 2006. Each rec change (or rating change) is an analyst's current rating minus his prior rating (prior ratings may be from 1993, but current ratings are from 1994 onward). Analyst initiations or ratings with no prior outstanding ratings are excluded. A rating is outstanding if it has been confirmed by the analyst (in the I/B/E/S review date field) in the last twelve months and has not been stopped by the broker (in the I/B/E/S Stopped File). Ratings are coded as 1 (sell) to strong buy (5), and rating changes lie between -4 and 4. Firms with less than three analysts making up the consensus are excluded. The chart plots the transition probabilities of rec changes—the probability that a prior rec transits to any of the five rating categories.

Read:

- Average recommendation returns are economically large in the full sample.
- Removing earnings announcements, management guidance, and multiple-recommendation days substantially reduces the mean CAR.
- Outlier removal further reduces the mean.
- Even after filtering, average effects remain statistically significant.

Economic message. Analysts often issue recommendation changes around firm news, so average event returns overstate the analyst's standalone information contribution. But the cleaned sample still contains a subset of recommendation changes with visible stock-level impact.

5.3 Table 3: how many recommendations are influential?

Read:

- 11.7% of recommendation changes are influential in returns.

- 12.8% are influential in turnover.
- 4.8% are influential under both definitions.
- 6.9% are influential in returns only, while 8.0% are influential in turnover only.

Economic message. Most analyst recommendation changes are not visibly influential. Still, the influential subset is large enough to matter and is not simply a few anecdotes.

5.4 Table 3: characteristics of influential recommendations

Characteristics	Influential based on firm's abnormal returns				Influential based on firm's abnormal turnover			
	Difference				Difference			
	Influ - Not				Influ - Not			
	Not Influ	Influential	Influ - Not	t-stat	Not Influ	Influential	Influ - Not	t-stat
Panel A: Analyst and recommendation characteristics								
Number of rec changes	77.537	10.292			76.595	11.234		
Forecast accuracy quintile	2.810	1.176	-0.639***	(-2.75)	2.810	12.8%	-0.634**	(-2.45)
Away from consensus	0.358	0.416	0.058***	(9.84)	0.355	0.434	0.079***	(14.15)
Star analyst	0.159	0.205	0.046***	(9.53)	0.160	0.198	0.038***	(8.19)
Absolute analyst experience (# Qtrs)	28.45	29.64	1.192***	(4.63)	28.44	29.61	1.169***	(4.62)
Relative analyst experience	2.652	3.354	0.702***	(3.03)	2.671	3.162	0.491***	(3.52)
Concurrent earnings forecast	0.446	0.510	0.064***	(10.15)	0.448	0.492	0.044***	(7.16)
Influential before (any stock)	0.562	0.664	0.102***	(15.57)	0.562	0.656	0.094***	(14.45)
Influential before (same stock)	0.100	0.137	0.037***	(9.99)	0.100	0.132	0.032***	(8.96)
Panel B: Firm characteristics prior to recommendation								
BM ratio	0.484	0.498	0.014***	(2.78)	0.485	0.485	-0.001	(-0.12)
Size (ln)	8071.7	7491.0	-1330.4***	(-5.48)	8010.9	8074.9	-136.03	(-0.47)
Institutional ownership	0.61	0.63	0.027***	(8.44)	0.61	0.64	0.029***	(9.69)
Dispersion $\times 100$	14.167	15.119	0.952	(1.31)	14.307	14.568	0.101	(0.15)
Idiosyncratic volatility $\times 100$	2.504	2.195	-0.310***	(-13.64)	2.501	2.340	-0.161***	(-11.28)
Total volatility $\times 100$	2.876	2.556	-0.319***	(-11.79)	2.872	2.807	-0.065***	(-5.56)
Daily turnover $\times 100$	0.655	0.603	-0.052***	(-7.42)	0.657	0.592	-0.065***	(-9.59)
Analyst activity (RPS forecasts)	86.669	72.422	-14.247***	(-15.56)	86.156	77.112	-9.044***	(-9.93)

(continued)

Characteristics	Influential based on firm's abnormal returns				Influential based on firm's abnormal turnover			
	Difference				Difference			
	Influ - Not				Influ - Not			
	Not Influ	Influential	Influ - Not	t-stat	Not Influ	Influential	Influ - Not	t-stat
Panel C: Change in firm environment around recommendation								
Leader-Follower Ratio (LFR) of rec	2.032	3.176	1.144***	(12.21)	1.955	3.619	1.664***	(16.57)
A Total volatility of daily ret $\times 100$	-0.081	0.380	0.431***	(21.41)	-0.087	0.218	0.265***	(13.60)
A Idiosyncratic volatility $\times 100$	-0.082	0.333	0.414***	(24.90)	-0.068	0.204	0.271***	(16.09)
A Daily turnover $\times 100$	0.076	0.375	-0.301	(-0.55)	0.834	0.601	-0.143	(-0.17)
A Analyst activity (# forecasts)	0.004	0.096	0.092***	(18.95)	-0.012	0.202	0.215***	(37.71)
A in FY1 (Forecast Revision) $\times 100$	-0.339	4.978	5.337***	(9.65)	-0.151	3.111	3.263***	(5.61)
A in FY2 (Forecast Revision) $\times 100$	0.034	0.127	0.093***	(3.93)	0.032	0.139	0.107***	(4.58)
A in LTG (Forecast Revision) $\times 100$	0.040	0.121	0.081***	(3.16)	0.035	0.156	0.122***	(4.25)
Fraction with large industry vw return	0.042	0.123	0.080***	(18.45)	0.048	0.079	0.031***	(9.17)
Fraction with large industry vw return	0.067	0.123	0.056***	(12.12)	0.071	0.093	0.022***	(5.65)

Influential (Influ) recommendations (recs) are compared with non-influential recs. Influential definitions are based either on prior returns or on prior abnormal turnover. For example, for returns, a rec change is influential if its correct-signed (0,1) CAR is 1.96 standard deviations greater than expected based on the firm's prior three-month idiosyncratic volatility of daily returns. The sample is from I/B/E/S (1984-2006) with earnings units, days, next forecast days, and multiple-rec days removed (sample 6 in Table 2). Restrictions are excluded. Panel A reports the average analyst or rec characteristic. Forecast accuracy quintile is the quintile rank (lower rank = greater accuracy) of the analyst based on his last submitted FY1 earnings forecast. A rec moves away from consensus (dummy) when the absolute deviation of the new rec from the consensus is larger than the absolute deviation of the prior rec from the consensus. Star analysts (dummy) are analysts who are ranked in Institutional Investor. Analyst experience is the no. of quarters since the analyst issued the first estimate or rec on I/B/E/S. Concurrent earnings forecast = 1 when the analyst issued any earnings forecast in the three-day window around the rec. Influential before indicates that the analyst was influential in the past. Panel B compares firm characteristics with variable definitions found in Section 3. Panel C summarizes the changes in the firm environment around the rec. Leader-Follower Ratio (LFR) is the gap sum of the prior two recs divided by the gap sum of the next two recs. A ratio > 1 indicates a leader analyst whose recs are quickly followed by other brokers' recs. We also compute the A in Total volatility of daily returns from the prior three months to the three months after the rec (including 2-5 days around the event). We also compute in the same manner other change variables. The reported values for Total, Idiosyncratic volatility, Dispersion, and Turnover are multiplied by 100 (e.g., 1.0 represents 1.0%). A Forecast Revision is computed for the FY1 and FY2 and long-term growth (LTG) consensus analyst forecast revisions from three months before to three months after the rec. FY1 and FY2 consensus forecast revisions are scaled by price (outlier revisions more than 100% of price are removed), and LTG forecasts are in percentages. Large Puma-French 49 industry returns = 1 when recommendations coincide with a large (> 1.96 index than expected) abnormal return movement in the firm's industry value-weighted (vw) or equal-weighted (ew) returns over days [0,1]. Asterisks in the difference columns indicate statistical significance using standard errors clustered by calendar day where *, **, and *** represent significance levels of 10%, 5%, and 1%, respectively.

Read:

- Influential calls are more likely to be away from consensus.
- Star analysts and previously influential analysts are overrepresented.
- Concurrent earnings forecasts are more common among influential calls.
- Experience differences are statistically significant, though not always the largest economically.

Economic message. Being influential is not random. It is connected to analyst reputation, previous success, and recommendations that challenge the consensus.

5.5 Table 3 Panel C: what happens after influential recommendations?

Table 1: Selected changes in the firm environment after influential recommendations

Variable	Non-influential	Influential	Difference
Leader-follower ratio	2.032	3.176	1.144***
Δ total volatility $\times 100$	-0.081	0.350	0.431***
Δ idiosyncratic volatility $\times 100$	-0.082	0.333	0.414***
Δ daily turnover $\times 100$	0.004	0.096	0.092***
Δ analyst activity	-0.359	4.978	5.337***
Δ FY2 forecast revision $\times 100$	0.034	0.127	0.093***
Large value-weighted industry return	0.042	0.123	0.080***

Read:

- Other analysts follow influential calls more quickly.
- Volatility and turnover rise after influential calls.

- Analyst activity and forecast-revision magnitudes increase.
- A larger share of influential calls coincides with large same-direction industry returns.

Economic message. Influential recommendations look like information events, not merely one-day price noise. They trigger broader activity by investors and other analysts.

5.6 Table 4: probit predicting influential recommendations

Table 4
Predicting when a recommendation change will be influential

Explanatory Variable	Influential based on firm's abnormal returns		Influential based on firm's abnormal turnover	
	Coefficient	Marg. Eff	Coefficient	Marg. Eff
Influential before (any stock)	0.154*** (8.35)	2.88%	0.130*** (7.17)	2.58%
Influential before (same stock)	0.065*** (2.93)	1.21%	0.052** (2.54)	1.03%
Rec level	0.045*** (4.01)	0.80%	0.008 (0.74)	0.16%
Absolute value of <i>recchg</i>	-0.017 (-1.00)	-0.16%	0.001 (0.03)	0.01%
Upgrade Dummy	0.080*** (4.01)	1.50%	-0.032 (-1.63)	-0.64%
Reg FD Dummy	0.206*** (8.82)	3.85%	0.219*** (9.24)	4.37%
Settlement Dummy	0.093*** (3.84)	1.73%	0.171*** (6.81)	3.42%
Past forecast accuracy quintile	-0.011* (-1.90)	-0.24%	-0.009 (-1.59)	-0.21%
Away from consensus	0.147*** (9.87)	2.74%	0.148*** (10.18)	2.96%
Star analyst	0.207*** (9.36)	3.87%	0.156*** (7.22)	3.11%
Absolute analyst experience	-0.001* (-1.96)	-0.41%	-0.001 (-1.61)	-0.31%
Relative analyst experience	0.001 (0.84)	0.14%	0.000 (0.36)	0.07%
Concurrent earnings forecast	0.119*** (7.87)	2.22%	0.110*** (7.71)	2.19%
Past Leader-Follower Ratio (LFR)	0.006*** (2.74)	0.36%	0.004** (1.98)	0.28%
Log(B/M)	-0.100*** (-9.66)	-1.51%	-0.061*** (-6.15)	-0.99%
Log(Size)	-0.082*** (-10.67)	-2.49%	0.017** (2.46)	0.53%
Price momentum	0.0331** (2.38)	0.34%	0.042*** (3.40)	0.49%
Log(Institutional ownership)	0.049** (2.15)	0.38%	0.138*** (5.30)	1.15%
Log(Turnover)	0.042*** (2.82)	0.62%	-0.084*** (-5.36)	-1.32%
Log(Idiosyncratic volatility)	-0.351*** (-15.10)	-3.45%	-0.090*** (-4.04)	-0.95%
Dispersion	0.032*** (3.13)	0.38%	0.033*** (2.92)	0.42%

(continued)

Table 4
Continued

Explanatory Variable	Influential based on firm's abnormal returns		Influential based on firm's abnormal turnover	
	Coefficient	Marg. Eff	Coefficient	Marg. Eff
Log(Analyst activity)	-0.138*** (-11.49)	-2.17%	-0.148*** (-12.86)	-2.48%
Pseudo R-sq	0.04745		0.03654	
# Observations	58384		58384	
Chi-Sq test	1485.68***		1323.57***	

The binary dependent variable is whether a recommendation (*rec*) is influential and the sample is sample 4 from Table 2 with reiterations excluded and firm and analyst characteristics required. Marginal effects are reported below the coefficient estimates. The marginal effect for continuous (dummy) explanatory variables represents the change in the predicted probability when the independent variable changes by one standard deviation (changes from 0 to 1). There are two definitions of influential. First, influential *recs* are those when a correct-signed CAR is 1.96 standard deviations greater than expected based on the firm's prior three-month *Idiosyncratic volatility* of daily returns. The second uses an abnormal turnover greater than 1.96 standard deviations of that expected from the abnormal daily turnover in the stock's prior three-month history. The *Rec Level* is the rating level after the *rec* change (*recchg*) (1 = sell to 5 = strong buy). The absolute value of the *recchg*, *Upgrade* dummy, *Reg FD* dummy (=1 after Aug 2000), and *Settlement* dummy (=1 in 2003 and after) are also included. Past *Forecast accuracy* quintile is the average quintile rank of the analyst (smaller ranks denote greater accuracy). *Away from consensus* = 1 when the absolute deviation of the new *rec* from the consensus is larger than the absolute deviation of the prior *rec* from the consensus. *Star analysts* = 1 for ranked analysts in the most recent *Institutional Investor* polls. *Absolute analyst experience* is measured as the # of quarters in I/B/E/S. *Relative analyst experience* subtracts the average experience of other covering analysts. *Concurrent earnings forecast* = 1 when the same analyst issued any earnings forecast in the three-day window around the *rec*. *Leader-Follower Ratios* larger than one denote leader analysts. *Turnover*, *Idiosyncratic volatility*, and *Dispersion* are based on prior three-month averages. *Analyst activity* is total number forecasts issued by all analysts in the prior three months. *, **, and *** denote significance levels of 10%, 5%, and 1%, respectively, using standard errors clustered in two dimensions (by analyst and firm) with *z* statistics in parentheses.

Read:

- Prior influence matters: being influential before for any stock raises the probability by 2.88 percentage points; prior influence for the same stock adds 1.21 points.
- Star analyst status has a large positive marginal effect of 3.87 percentage points.
- Away-from-consensus recommendations and concurrent earnings forecasts have large positive effects.
- Reg FD and the Global Analyst Settlement are associated with higher probabilities of influential recommendations.
- Size, idiosyncratic volatility, and analyst activity lower the probability of an influential return event.

Economic message. The high-impact subset is predictable. It comes from recognized analysts, non-consensus calls, recommendations with additional forecast information, and firms where a new interpretation can more plausibly change beliefs.

5.7 Table 5: ever-influential versus never-influential analysts

Read:

- About one quarter of analysts never issue a return-influential recommendation in the sample.
- Conditional on ever being influential, the typical analyst has 22.1% of recommendations classified as influential.
- Ever-influential analysts are more experienced and more likely to have been star analysts.

Economic message. Influence is concentrated. Some analysts repeatedly produce visible calls, while a nontrivial fraction never does.

Table 5
Analysts who had at least one influential recommendation change

Characteristics	Influential based on firm's abnormal returns				Influential based on firm's abnormal turnover			
	Difference		t-stat		Difference		t-stat	
	Never	Ever-Influential			Never	Ever-Influential		
Number of Analysts	935	2,821			781	2,975		
% influential recs for typical analyst	0.0%	22.1%			0.0%	23.5%		
Forecast accuracy quintile	2.894	2.819			2.907	2.819		
Away from consensus	0.357	0.367			0.010	0.348		
Was once a Star analyst	0.092	0.253			0.161***	0.101		
Absolute analyst experience (# Qtrs)	18.09	24.42			6.329***	18.71		
Relative analyst experience	-1.26	1.36			2.624***	0.63		
Concurrent earnings forecast	0.477	0.462			-0.013*	0.467		

This table reports the average analyst or recommendation (rec) characteristic by analysts who were ever influential versus those who were never influential. The percentage of analyst recs that are influential is the average proportion of an individual analyst's recs that are influential conditional on the analyst ever being influential in the whole sample. Only analysts who had at least 5 recommendations (recs) in the 1994-2006 period are included. There are two definitions of influential. First, influential recs are those when a correct-signed [0,1] CAR is 1.96 standard deviations greater than expected based on the firm's prior three-month *idiosyncratic* volatility of daily returns. The second uses an abnormal turnover greater than 1.96 standard deviations of that expected from the abnormal daily turnover in the stock's prior three-month history. The sample is from 1982/8 with earnings announcement days, management forecast days, and multiple-rec days removed (sample 4 in Table 2). 1982/8 rec dates are replaced by those from First Call whenever available. Reiterations (rec change = 0) are also excluded. Forecast accuracy quintile is the quintile rank of the analyst based on his last unreviewed FY1 earnings forecast for that fiscal year according to Loh and Meier (2006). A rec moves Away from consensus (dummy variable) when the absolute deviation of the new rec from the consensus is larger than the absolute deviation of the prior rec from the consensus (as in Jegadeesh and Kim 2010). *Star analyst* (dummy variable) are analysts who are ranked as All-Americans in the most recent annual Institutional Investor poll. *Absolute analyst experience* is measured as the # of quarters in 1982/8. *Relative analyst experience* subtracts the average experience of other covering analysts. *Concurrent earnings forecast* is a dummy indicating whether the analyst issued any type of earnings forecast in the three-day window around the rec. Asterisks in the difference column indicate statistical significance using standard errors clustered by calendar day, where *, **, and *** represent significance levels of 10%, 5%, and 1%, respectively.

Table 6
Different definitions of influential

Method	Using alternative methods to define influential	Total	Non-Influential	Influential	Percentage
0	Recchg Original	87,829	77,537	10,292	11.7%
1	Recchg_Last	50,024	44,932	5,092	10.2%
2	Recchg_Consensus	96,282	86,942	9,340	9.7%
3	Influential based on raw returns	87,829	78,379	9,450	10.8%
4	Using rec-free days to compute prior volatilities	87,829	77,203	10,626	12.1%
5	Removing obs with large pre-event absolute return	76,638	68,100	8,538	11.1%

The fraction of influential recommendation changes (*recchg*) for different subsamples (methods) is reported. The definition of influential here is based on abnormal returns. The original *recchg* (method 0) variable is the analyst's current rec minus the analyst's own prior rec. A prior rec is outstanding only if it has a less-than-one-year-old review date and has not been stopped by the broker. A *recchg* is considered influential when the [0,1] CAR is the same sign as the *recchg* and the CAR is 1.96 standard deviations greater than expected based on the firm's prior three-month *idiosyncratic* volatility of daily returns. Reiterations are excluded since the *recchg* is basically zero. Methods 1 and 2 amend the *recchg* definition: 1 is based on the current rec minus the last rec from any analysts (*recchg-last*), and 2 is based on the current rec minus the most recent consensus (*recchg-con*). Method 3 is method 0 except that a *recchg* is influential if the cumulative raw (not abnormal) return is influential. Method 4 amends method 0 by using only rec-free days to compute the prior return volatility of the firm. This removes days with recommendations from the prior three-month [-69, -6] period used to compute *idiosyncratic* volatility. Method 5 uses method 0 except that it removes observations where the pre-event [-2, -1] contains an influential abnormal return in any direction.

5.8 Table 6: robustness to alternative definitions

Table 2: Alternative definitions of influential in returns

Method	Total	Influential	Percentage
Original recommendation change	87,829	10,292	11.7%
Current rec minus last rec by any analyst	50,024	5,092	10.2%
Current rec minus consensus rec	96,282	9,340	9.7%
Influential based on raw returns	87,829	9,450	10.8%
Use rec-free days to compute prior volatility	87,829	10,626	12.1%
Remove large pre-event absolute returns	76,638	8,538	11.1%

Read:

- The influential fraction stays close to the baseline under several alternative rules.
- Raw returns produce 10.8%, not a higher number, so the result is not created by benchmark adjustment.
- Removing large pre-event returns still leaves 11.1% influential recommendations.

Economic message. The main result is robust: about one out of ten cleaned recommendation changes has a visible stock-level effect.

Table 7
Influential fraction of earnings forecasts with and without stock recommendations

Panel A: Forecast revisions sample							
Forecast revision sample	Influential based on abnormal:	All forecast revisions			Revisions with recommendations		
		Not Influ	Influential	Percent	Not Influ	Influential	Percent
Annual	Returns	286,813	13,402	4.5%	18,094	2,023	10.1%
	Turnover	283,672	16,543	5.5%	17,676	2,441	12.1%
Quarterly	Returns	105,570	5,346	4.8%	6,268	799	11.3%
	Turnover	104,125	6,791	6.1%	6,081	986	14.0%
LTG forecasts	Returns	42,258	1,750	4.0%	3,119	310	9.0%
	Turnover	41,055	2,953	6.7%	2,930	499	14.6%
Panel B: Recommendation changes sample							
Sample	Influential based on abnormal:	All recommendation changes			Rec changes with earnings forecasts		
		Not Influ	Influential	Percent	Not Influ	Influential	Percent
Rec Change	Returns	77,537	10,292	11.7%	34,608	5,253	13.2%
	Turnover	76,595	11,234	12.8%	34,331	5,530	13.9%

We examine what fraction of earnings forecasts revisions are influential (Influ) according to our definition. Earnings forecasts revisions based on yearly (FY1), quarterly (Q1), and long-term-growth (LTG) horizons are from the I/B/E/S Detail file from 1994 to 2006 where a revision is coded as an upward or downward revision based the current forecast minus the prior outstanding forecast by the same analyst. Dates from First Call are used as revision dates whenever available through hand-matching of broker name. The samples are then screened for contaminating events based on the sequence shown in Table 2. The final sample considered here is based on forecast revisions without contemporaneous earnings announcements, company guidance, and multiple same-horizon forecasts days. A revision is considered influential if its two-day event [0,1] return reaction is in the correct direction and 1.96 standard deviations more than expected based on the prior three-month daily *Idiosyncratic volatility* of stock returns. Abnormal turnover is also used as a benchmark for being influential. An earnings forecast is treated as accompanied by a recommendation when there is a recommendation issued by the same analyst in the three-day window around the earnings forecast revision date. Panel B, we list influential fractions based on the original recommendation-change sample in Table 4.

5.9 Table 7: recommendation changes versus earnings forecast revisions

Read:

- Standalone earnings forecast revisions are influential only about 4–6% of the time.
- Forecast revisions paired with recommendation changes become much more influential.
- Recommendation changes are influential more often than standalone earnings forecast revisions.

Economic message. The recommendation is the analyst output that most clearly communicates a market-relevant view. Earnings forecasts are more powerful when bundled with a recommendation change.

6 Short summary

- The paper asks when analyst recommendation changes visibly affect individual stocks.
- It criticizes the standard average-event-study approach because a significant average effect does not imply a noticeable firm-level effect.
- It defines a recommendation change as influential in returns if the event CAR is correct-signed and exceeds a firm-specific volatility threshold.
- It also defines influential in turnover using abnormal trading volume.
- After removing firm-news-contaminated days, only 11.7% of recommendation changes are influential in returns and 12.8% in turnover.
- Influential calls are more likely from star, leader, and previously influential analysts.
- Influential calls are more likely when the recommendation moves away from consensus or is accompanied by an earnings forecast.
- Influential calls are more likely for firms where beliefs are harder to pin down: growth firms, smaller firms, high institutional ownership firms, high dispersion firms, and lower analyst activity firms.

- Influential calls are followed by higher turnover, higher volatility, more analyst activity, larger forecast revisions, and industry return spillovers.
- The main interpretation is a paradigm shift: some analyst calls change how investors and analysts assess the firm.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that analyst recommendation changes are not uniformly informative. Most recommendation changes do not visibly affect the stock at the firm level, but a meaningful minority do. The influential subset is predictable from analyst reputation, prior influence, recommendation content, and firm information environment.

7.2 Economic intuition

An analyst recommendation matters when it changes the model investors use to think about a firm. Calls that are away from consensus, made by analysts with reputational capital or a history of influence, and issued for firms with uncertain valuations are more likely to generate such a change. This is why influential recommendations are followed by broader analyst activity, volatility, turnover, and forecast revisions.

7.3 Takeaway

The value of sell-side research is concentrated in rare but visible calls. Average stock-price reactions are too blunt to capture this. The right question is not only whether recommendations move prices on average, but which recommendations are strong enough to be noticed by investors following the stock.